

1 考点 1:Measures of Financial Risk

1. A quantitative analyst at a cryptocurrency trading firm is developing a risk model for digital assets. During a team meeting, she needs to explain why traditional risk models based on normal distribution might not be suitable for crypto assets. Which of the following statements about the return distributions of cryptocurrency investments is most likely correct?

- A. Machine learning models can accurately capture cryptocurrency returns by focusing solely on the mean return, as other parameters are less significant in digital asset markets.
- B. The return distributions of cryptocurrency investments typically show thinner tails than traditional assets, making extreme price movements less frequent than in conventional markets.
- C. Cryptocurrency returns tend to exhibit more pronounced peaks and fatter tails than the normal distribution would suggest, particularly during periods of market stress.
- D. When applying normal distribution models to estimate potential losses in cryptocurrency trading, the probability of extreme events is typically overestimated compared to actual market behavior.

答案:C

解析:

A 选项错误。

- 即使使用先进的机器学习模型，仅依靠均值也无法充分描述加密货币收益的分布特征。收益分布的完整描述需要多个参数，包括但不限于均值、标准差、偏度和峰度。

B 选项错误。

- 加密货币市场实际上表现出比传统资产更为显著的肥尾特征，极端价格波动的频率远高于正态分布预期，这反映了数字资产市场的高波动性特征。

C 选项正确。

- 加密货币收益分布通常表现出显著的高峰肥尾特征，这意味着小幅波动和极端事件的发生频率都高于正态分布预期。这种特征在市场压力期间尤为明显，反映了数字资产市场的独特风险特征。

D 选项错误。

- 使用正态分布模型估计加密货币市场的极端损失时，实际上会低估而非高估极端事件的概率，这是由于实际市场中肥尾现象的存在。

考察: 对比不同金融资产的收益率分布。（高峰肥尾现象）

2. A financial analyst evaluates a trading portfolio and determines that the Value at Risk (VaR) at a 95% confidence interval for a 1-day holding period is \$500,000. Which of the following statements is TRUE?

- A. The probability of the daily loss on the portfolio exceeding \$500,000 is 5%.
- B. There is a 95% confidence that the daily loss on the portfolio will not exceed \$500,000.
- C. The maximum daily loss that the portfolio can incur is \$500,000.
- D. 95% of financial analysts agree that the maximum loss on the portfolio is \$500,000.

答案:B

解析:

A 选项错误。

- VaR 的定义是，在 95% 的置信水平下，损失不会超过 \$500,000。这意味着分析师有 95% 的信心认为损失不会超过这个值，而不是说损失发生的概率就是 5%。

B 选项正确。

- VaR 表示在 95% 的置信水平下，投资组合的每日损失不会超过 \$500,000。这是 VaR 的核心定义，因此选项 B 是正确的。

C 选项错误。

- VaR 并不代表投资组合可能遭受的最大损失。它只是一个在特定置信水平下的估计，投资组合可能会遭受超过 \$500,000 的损失，但这种情况发生的概率较低。

D 选项错误。

- VaR 是基于统计计算的风险度量，与金融分析师的主观意见无关。95% 的金融分析师是否同意这一点并不影响 VaR 的定义。

考察: VaR 的定义（置信就是有信心认为）

3. A risk management team is discussing the implications of Value at Risk (VaR) for their investment strategies. Which of the following statements about VaR is NOT correct?

- A. VaR can account for the diversified holdings of a financial institution, reducing capital requirements.
- B. $\text{VaR}(10\%) = \$0$ indicates that a positive dollar return is likely to occur on 90 out of 100 days.
- C. $\text{VaR}(1\%)$ can be interpreted as the number of days in which the portfolio value will exceed a loss of 1%.
- D. A major limitation of the VaR measure is the use of two arbitrary parameters: confidence level and holding period.

答案:C

解析:

A 选项正确。

- VaR 确实可以考虑金融机构投资组合的多样化，从而帮助确定在一定置信水平下，投资组合在特定时间内可能遭受的最大损失。这有助于金融机构更有效地分配资本，以覆盖潜在的风险。

B 选项正确。

- $\text{VaR}(10\%) = \$0$ 表示在 90% 的置信水平下，100 天中有 10 天可能会出现损失超过 0 美元的情况。换句话说，有 90% 的概率损失不会超过 0 美元，而不是说有 90 天会出现正的回报。但是题目说的是“indicates”，也有一种“可能”的意思，因此，选项 B 可以被视为正确。

C 选项错误。

- $\text{VaR}(1\%)$ 表示在 99% 的置信水平下，损失超过 1% 的情况在 100 天中可能发生 1 天。它并不直接表示损失超过 1% 的具体天数，而是表示在给定的置信水平下，损失超过 1% 的最大可能天数。因此，选项 C 的描述是不准确的。

D 选项正确。

- VaR 的一个主要限制是它依赖于两个任意参数：置信水平和持有期。VaR 的结果可能会因这些参数的选择而有所不同，限制了其在不同情境下的适用性和可靠性。
- 这些参数的选择会显著影响 VaR 的计算结果。例如，较高的置信水平（如 99%）会导致更高的 VaR 值，从而可能导致过度保守的资本要求；而较低的置信水平（如 90%）则可能低估潜在风险。
- 此外，持有期的选择也会影响 VaR 的结果。短期持有期可能无法充分反映长期投资的风险特征。

考察: VaR 的特点。（VaR 的一个主要限制是它依赖于两个任意参数：置信水平和持有期）

4. When using the delta-normal method to calculate Value-at-Risk (VaR) for a linear portfolio, how would the presence of fat tails in the return distribution affect the estimated VaR?

- A. The estimated VaR would likely underestimate the actual VaR.
- B. The estimated VaR would be equal to the actual VaR.
- C. The estimated VaR would likely overestimate the actual VaR.

D. The estimated VaR cannot be accurately determined without additional information.

答案:A

解析:

A 选项正确。

- 在存在肥尾现象的情况下，收益分布的尾部损失发生频率较高。使用 delta-normal 方法计算的 VaR 通常假设收益分布是正态的，这会导致对潜在极端损失的低估。因此，实际的 VaR 可能会高于估计的 VaR，选项 A 的描述是正确的。

B 选项错误。

- 由于肥尾现象的存在，估计的 VaR 不太可能与实际 VaR 相等，因此选项 B 是错误的。

C 选项错误。

- 由于肥尾现象，估计的 VaR 通常会低于实际 VaR，而不是高于，因此选项 C 是错误的。

D 选项错误。

- 肥尾会导致对 VaR 的低估，因为它忽略了肥尾对 VaR 的既定影响。因此，这个选项是不正确的。

考察: 肥尾现象对 VaR 的影响。（肥尾会导致对 VaR 的低估）

5. A portfolio manager at a hedge fund is conducting a training session on risk measurement principles. While explaining the concept of consistent risk measures, she notices that some junior analysts have misconceptions about these measures. Which of the following statements represents a misconception about consistent risk measures?

- A. Monotonicity implies that if portfolio A consistently outperforms portfolio B, then $\text{Risk}(A) < \text{Risk}(B)$.
- B. Subadditivity indicates that the risk of two combined portfolios will not exceed the sum of individual risks, i.e., $\text{Risk}(A+B) \leq \text{Risk}(A) + \text{Risk}(B)$.
- C. Homogeneity means that if the scale of portfolio A doubles, its risk measure should also double, i.e., $\text{Risk}(2A) = 2\text{Risk}(A)$.
- D. Translation invariance implies that adding cash to a portfolio reduces its risk measure, but the reduction is not necessarily equal to the cash added.

答案:D

解析:

A 选项正确，不选。

- 单调性确保风险度量与投资组合的表现一致，表现差的投资组合具有更高的风险度量。

B 选项正确，不选。

- 次可加性确保组合的风险不会合理地增加。

C 选项正确，不选。

- 齐次性确保风险度量与投资规模成比例，这符合风险度量的直观理解。

D 选项错误。

- 根据平移不变性的定义，向投资组合中添加现金 K 应使风险度量精确减少 K，而不是可能不相等。这是风险度量的一个关键性质，确保添加无风险资产会直接减少风险度量。

考察: 风险度量的性质。（齐次性、次可加性、平移不变性）

6. An investor has purchased a stock that is part of the S&P 500 index. Considering the statistical characteristics of stock returns, which of the following traits is most likely to be exhibited by the returns of this stock compared to a normal distribution?

- A. The return distribution has low kurtosis, indicating a shape that is flatter than a normal distribution.

- B. The return distribution is symmetric around zero, meaning the left and right sides of the distribution are balanced.
- C. The return distribution lacks excess kurtosis, meaning the tails are similar to those of a normal distribution, with few extreme values.
- D. The return distribution is more extreme, with more extreme values likely to occur in both positive and negative directions.

答案:D

解析:

A 选项错误。

- 实际上，股票回报率的分布往往比正态分布更尖峰（更高的峰度），意味着大多数回报率集中在均值附近，分布的形状比正态分布更为尖锐。

B 选项错误。

- 股票回报率的分布通常表现出负偏度，即分布的左侧尾部更长，表明在不利市场条件下可能出现更大的损失。

C 选项错误。

- 股票回报率的分布通常具有超额峰度，即分布的尾部比正态分布更厚，这意味着极端值（无论是正向还是负向的极端回报）出现的概率更高。

D 选项正确。

- 股票回报率的分布倾向于具有更厚的尾部，这表明在正向和负向都有可能出现更多的极端值。这是由于金融市场的波动性和极端事件的潜在影响。

考察: 股票回报率的分布特征。（尖峰、负偏度、超额峰度）

7. A risk manager at a global investment bank is evaluating different risk metrics for their trading portfolio. The team is debating whether to use Value at Risk (VaR) or Expected Shortfall (ES) as their primary risk measure for regulatory reporting and internal risk control.Which of the following statements about these two risk measures is correct?

- A. Expected Shortfall satisfies the portfolio diversification principle (sub-additivity), while VaR may show inconsistent results when aggregating portfolio risks.
- B. VaR and Expected Shortfall provide identical risk information since they both measure potential losses over a specified time horizon at a given confidence level.
- C. Both VaR and Expected Shortfall require the portfolio returns to follow a normal distribution to be valid risk measures.
- D. Expected Shortfall remains constant regardless of the confidence level chosen, while VaR varies with different confidence levels.

答案:A

解析:

A 选项正确。

- Expected Shortfall 满足风险度量的子可加性原则，这意味着分散化投资总能降低组合风险。而 VaR 可能出现两个子组合的风险之和小于组合总风险的异常情况，这违反了投资组合理论中分散化降低风险的基本原则。

B 选项错误。

- 虽然两个指标都衡量潜在损失，但它们提供的信息不同。VaR 只表示特定置信水平下的最大损失值，而 Expected Shortfall 则进一步告诉我们超过 VaR 时的平均损失大小，提供了更完整的尾部风险信息。

C 选项错误。

- 这两种风险度量方法都不要求收益必须服从正态分布。它们可以应用于任何类型的收益分布，包括偏态分布和厚尾分布。正态分布假设只是为了简化计算过程。

D 选项错误。

- Expected Shortfall 和 VaR 都会随着置信水平的变化而变化。提高置信水平会导致 VaR 值增加，相应的 Expected Shortfall 也会随之改变，因为它是基于 VaR 以外损失的条件期望。

考察: 风险度量指标比较（重点是 VaR 和 ES 的特性差异）

8. A risk management team at a global investment firm is evaluating different risk measures to enhance their portfolio management strategies. During a meeting, the team discusses the advantages of expected shortfall (ES) over value at risk (VaR) in capturing tail risk. Which of the following is not a reason that expected shortfall (ES) is a more appropriate risk measure than value at risk (VaR)?

- A. For normal distributions, only ES satisfies all the properties of coherent risk measurements.
- B. For nonelliptical distributions, the portfolio risk surface formed by holding period and confidence level is more convex for ES.
- C. ES gives an estimate of the magnitude of a loss.
- D. ES has less restrictive assumptions regarding risk/return decision rules than VaR.

答案:A

解析:

A 选项错误。

- VaR 和 ES 在正态分布下都满足一致性风险度量的所有性质。然而，当不满足正态性假设时，只有 ES 满足所有一致性风险度量的性质。因此，选项 A 的描述是不正确的。

B 选项正确。

- 对于非椭圆分布，ES 形成的投资组合风险曲面在持有期和置信水平上更凸。这意味着 ES 在处理复杂分布时提供了更好的风险评估。

C 选项正确。

- ES 提供了损失幅度的估计，而不仅仅是一个置信水平下的最大损失值。这使得 ES 在评估极端损失时更为有效。

D 选项正确。

- ES 在风险/收益决策规则方面的假设较少限制性，这使得它在不同市场条件下更具适用性。

考察: ES 和 VaR 的特点。（ES 在非正态分布中也满足一致性风险度量）

9. A financial institution has determined that the Value-at-Risk (VaR) for a portfolio over a 1-day horizon is USD 200 million. Based on this information, which of the following statements is TRUE regarding the VaR over a 5-day horizon?

- A. The VaR over a 5-day horizon is USD 447 million if returns are independently and identically distributed.
- B. The VaR over a 5-day horizon is USD 447 million if returns are not independently and identically distributed.
- C. The VaR over a 5-day horizon remains USD 200 million since VaR is independent of the time horizon.
- D. The VaR over a 5-day horizon is USD 40 million, regardless of the distribution of returns.

答案:A

解析:

A 选项正确。

- 根据时间平方根法则，假设收益是独立同分布的，5 天的 VaR 可以通过 1 天的 VaR 乘以 $\sqrt{5}$ 来计算。具体计算如下：

$$\text{VaR}_{5 \text{ days}} = \text{VaR}_{1 \text{ day}} \times \sqrt{5} = 200 \text{ million} \times \sqrt{5} \approx 200 \text{ million} \times 2.236 = 447.21 \text{ million}$$

B 选项错误。

- 虽然计算结果为 447 百万美元，但如果收益不是独立同分布的，使用时间平方根法则不再适用。这意味着在这种情况下，VaR 的计算可能会低估或高估实际风险，因此选项 B 不正确。

C 选项错误。

- VaR 确实依赖于时间范围。1 天的 VaR 不能简单地认为在 5 天内保持不变，因为风险会随着时间的延长而增加。

D 选项错误。

- USD 40 million 的计算没有依据，且未能反映 VaR 的实际风险水平。此选项没有考虑到 VaR 的计算方法和时间平方根法则。

考察: 时间平方根法则。（平方根法则前提是独立同分布）

10. A risk analyst at a commodity trading firm is evaluating an options portfolio that has a two-day Value-at-Risk (VaR) of \$3 million. The analyst needs to determine the appropriate translation of this VaR to a ten-day horizon under normal market conditions. What would be the estimated VaR for the ten-day horizon?

- A. \$4.743 million
- B. \$5.678 million
- C. \$6.708 million
- D. \$7.890 million

答案:C

解析:

步骤 1: 确定 2 天的 VaR。

- 根据题目，已知 2 天的 VaR 为：

$$\text{VaR}(2 \text{ days}) = 3,000,000$$

步骤 2: 推导 1 天的 VaR。

- 使用时间平方根法则，VaR 与时间的平方根成正比。根据理论公式：

$$\text{VaR}(n \text{ days}) = \text{VaR}(1 \text{ day}) \times \sqrt{n}$$

- 对于 2 天的 VaR，可以表示为：

$$\text{VaR}(2 \text{ days}) = \text{VaR}(1 \text{ day}) \times \sqrt{2}$$

- 由此可以推导出 1 天的 VaR：

$$\text{VaR}(1 \text{ day}) = \frac{\text{VaR}(2 \text{ days})}{\sqrt{2}} = \frac{3,000,000}{\sqrt{2}} \approx 3,000,000 \times 0.707 \approx 2,121,320$$

步骤 3: 计算 10 天的 VaR。

- 然后，计算从 1 天到 10 天的转换因子：

$$\text{因子}(1 \text{ 到 } 10 \text{ 天}) = \sqrt{10} \approx 3.162$$

- 将 1 天的 VaR 与计算得到的因子相乘，得到 10 天的 VaR：

$$\text{VaR}(10 \text{ days}) = \text{VaR}(1 \text{ day}) \times \text{因子}(1 \text{ 到 } 10 \text{ 天})$$

$$\text{VaR}(10 \text{ days}) = 2,121,320 \times 3.162 \approx 6,708,000$$

考察: 时间平方根法则。（需要知道转换因子的思维）

11. A risk analyst at a major investment bank is preparing a training session for new employees about Value at Risk (VaR). During the preparation, she reviews several statements about VaR’s characteristics and interpretations. Which of the following statements about VaR is CORRECT?

- A. VaR calculations primarily focus on individual asset risks, with portfolio diversification effects having minimal impact on capital requirements for financial institutions.
- B. When a portfolio's VaR(10%) equals \$0, this indicates the portfolio will maintain positive returns for 90 consecutive trading days.
- C. VaR was originally developed by commercial banks as a tool to better estimate their economic capital requirements while accounting for portfolio diversification effects.
- D. VaR measures provide precise predictions of future portfolio losses through advanced statistical modeling techniques.

答案:C

解析:

A 选项错误。

- VaR 计算实际上充分考虑了投资组合的多样化效应，这是其核心优势之一。通过计入资产间的相关性，VaR 能够反映分散化带来的风险降低效果，从而可能降低金融机构的资本要求。

B 选项错误。

- VaR(10%) = \$0 的含义是在任何给定日期有 90% 的概率获得正收益，而不是连续 90 天保持正收益。这种解释混淆了概率特征和时间序列的概念。

C 选项正确。

- VaR 确实最初是由商业银行开发，目的是为了在考虑投资组合多样化效应的情况下，更准确地估计经济资本需求。这种方法帮助银行更好地理解和管理其风险敞口。

D 选项错误。

- VaR 是一个统计概率工具，提供的是在特定置信水平下的潜在损失估计，而不是精确的损失预测。它的结果受到多个因素的影响，包括模型假设、数据质量和市场条件等。

考察: VaR 的基本特性和解释（重点是 VaR 的概率解释和应用目的）

12. A portfolio manager at a global investment bank is conducting a training session for junior risk analysts. During the session, she notices that some analysts have misconceptions about Value at Risk (VaR). Which of the following statements represents a common misconception about VaR?

- A. A 95% VaR of \$10 million means there is a 5% chance that the portfolio losses will exceed \$10 million over the specified time horizon.
- B. VaR can be used to compare risk levels across different trading desks, as it provides a standardized risk measure in monetary terms.
- C. When calculating portfolio VaR, correlation effects between different assets are considered, which typically results in a portfolio VaR less than the sum of individual VaRs.
- D. If a trading desk reports a daily 99% VaR of \$1 million, it means the desk will lose exactly \$1 million on its worst trading days.

答案:D

解析:

A 选项正确。

- 这是 VaR 置信水平的准确统计解释。95% 的置信水平意味着在 95% 的情况下损失不会超过 VaR 值，而 5% 的显著性水平 (α) 表示超过 VaR 值的概率。需要注意的是，时间维度（如日 VaR、周 VaR）的明确指定很重要，因为风险暴露会随时间变化。

B 选项正确。

- VaR 确实提供了一个标准化的风险度量，可以用货币单位表示不同交易部门的风险水平。这种统一的度量方式使得风险管理者能够比较不同资产类别（如股票与债券）的风险水平，便于设定风险限额和进行资本分配，同时也有助于高层管理者了解整体风险状况。

C 选项正确。

- 这反映了投资组合理论的核心原则。当资产间的相关系数小于 1 时，可以获得多样化收益。组合 VaR 的计算需要考虑完整的方差-协方差矩阵，因此通常小于单个资产 VaR 的简单加总。例如，两个负相关资产的组合可以显著降低整体 VaR 水平。

D 选项错误。

- 这个说法混淆了 VaR 的基本概念。VaR 是统计意义上的最小损失值，而不是确切的损失预测。99% 的置信水平意味着在 1% 的情况下会出现超过 VaR 的损失，而这些损失可能远超 VaR 值。这也是为什么在实践中需要补充使用压力测试等其他风险管理工具。

考察： VaR 的基本概念（重点是统计解释、实际应用及其局限性）

13. A risk analyst at a fixed-income investment firm is currently evaluating the potential risks associated with a bond position of \$15 million. The analyst notes that the modified duration of the bond is 4.2 years, and the annualized yield volatility is 1.5%. Using the duration method and assuming that the daily return on the bond position is independently identically normally distributed, the analyst needs to calculate the 10-day holding period VaR of the position with a 99% confidence interval, assuming there are 252 business days in a year.

- A. \$450,123
- B. \$425,678
- C. \$392,456
- D. \$367,890

答案:D

解析：

步骤 1：计算每日 VaR。

$$\text{VaR}(\text{dy}) = Z \times \sigma = 2.33 \times \frac{0.015}{\sqrt{252}} \approx 0.0092$$

其中，Z 为 99% 置信区间的 Z 值（约为 2.33），σ 为年化收益波动率。年化收益波动率的计算公式为：

$$\sigma = \text{Annualized Volatility} = \frac{\text{Daily Volatility}}{\sqrt{T}}$$

这里，T 是每年的交易日数量。

步骤 2：计算 10 天的 VaR。

$$\text{VaR}(10 \text{ days}) = \text{Modified Duration} \times \text{Bond Position} \times \text{VaR}(\text{dy})$$

$$\text{VaR}(10 \text{ days}) = 4.2 \times 15,000,000 \times 0.0092 \approx 367,890$$

因此，选项 D 的描述是正确的，10 天的 VaR 为 \$367,890。

考察： VaR 的计算。（注意年化波动率计算）

14. Mixed Fund has a portfolio worth USD 12,428,000 that consists of 42% of fixed income investments and 58% of equity investments. The 95% annual VaR for the entire portfolio is USD 1,367,000 and the 95% annual VaR for the equity portion of the portfolio is USD 1,153,000. Assume that there are 250 trading days in a year and that the correlation between stocks and bonds is zero. What is the 95% daily VaR for the fixed income portion of the portfolio?

- A. USD 21,263
- B. USD 46,445

C. USD 55,171

D. USD 72,635

答案:1 B

解析:

步骤 1：确定已知数据和假设，理解对应的公式。

- 投资组合的方差：

$$\text{Var}(\text{Portfolio}) = \text{Var}(A) + \text{Var}(B)$$

如果资产 A 和资产 B 是独立的，则投资组合的方差可以表示为各个资产方差的和。

- VaR 与方差之间的关系可以表示为：

$$\text{VaR} = k \cdot \sqrt{\text{Var}}$$

- 组合 VaR 的推导：

- 将方差的表达式代入 VaR 的公式中：

$$\text{VaR}_{\text{portfolio}} = k \cdot \sqrt{\text{Var}(A) + \text{Var}(B)}$$

- 由于 $\text{Var}(A) = \left(\frac{\text{VaR}_A}{k}\right)^2$ 和 $\text{Var}(B) = \left(\frac{\text{VaR}_B}{k}\right)^2$ ，我们可以得到：

$$\text{VaR}_{\text{portfolio}} = k \cdot \sqrt{\text{Var}(A) + \text{Var}(B)}$$

代入方差的表达式：

$$\text{VaR}_{\text{portfolio}} = k \cdot \sqrt{\left(\frac{\text{VaR}_A}{k}\right)^2 + \left(\frac{\text{VaR}_B}{k}\right)^2}$$

简化后得到：

$$\text{VaR}_{\text{portfolio}} = \sqrt{\text{VaR}_A^2 + \text{VaR}_B^2}$$

步骤 2：计算固定收益部分的 VaR。

根据 VaR 的平方和公式：

$$\text{VaR}_{\text{portfolio}}^2 = \text{VaR}_{\text{equity}}^2 + \text{VaR}_{\text{fixed income}}^2$$

将已知值代入公式：

$$1,367,000^2 = 1,153,000^2 + \text{VaR}_{\text{fixed income}}^2$$

计算：

$$1,367,000^2 = 1,868,689,000,000$$

$$1,153,000^2 = 1,327,609,000,000$$

代入计算：

$$1,868,689,000,000 = 1,327,609,000,000 + \text{VaR}_{\text{fixed income}}^2$$

解出固定收益部分的 VaR 平方：

$$\text{VaR}_{\text{fixed income}}^2 = 1,868,689,000,000 - 1,327,609,000,000 = 541,080,000,000$$

取平方根得到固定收益部分的年 VaR：

$$\text{VaR}_{\text{fixed income}} = \sqrt{541,080,000,000} \approx 734,357$$

步骤 3：将年 VaR 转换为日 VaR。

使用时间平方根法则将年 VaR 转换为日 VaR：

$$\text{VaR}_{\text{daily}} = \frac{\text{VaR}_{\text{fixed income}}}{\sqrt{250}} = \frac{734,357}{\sqrt{250}} \approx 46,445$$

考察: VaR 的计算。（计算组合 VaR）

15. Greenfield Investments is estimating a daily VaR for its equity portfolio currently valued at USD 1.2 billion. Using returns for the last 500 days (ordered in decreasing order, from highest daily return to lowest daily return), the daily returns are the following: 2.10%, 1.95%, 1.85%, 1.80%, ..., -1.75%, -1.90%, -1.95%, -2.05%, -2.15%. At the 99% confidence level, what is your estimate of the daily VaR using the historical simulation method?

- A. USD 24.00 million
- B. USD 22.50 million
- C. USD 20.00 million
- D. USD 25.50 million

答案:A

解析:

要估算 99% 置信水平下的每日 VaR，我们需要确定对应于最差 1% 收益的回报。

步骤 1：确定观察数量。

- 由于有 500 天的收益数据，最差 1% 对应于最低的 5 个收益（因为 $500 \times 0.01 = 5$ ）。

步骤 2：识别第 5 个最低收益。

- 从排序的收益中，第 5 个最低收益为-2.05%。

步骤 3：计算 VaR。

- VaR 的计算公式为：

$$\text{VaR} = \text{投资组合价值} \times \text{收益率}$$

- 代入数值：

$$\text{VaR} = 1,200,000,000 \times (-2.05\%) = -24,600,000$$

- 由于 VaR 通常以正数表示，我们取其绝对值：

$$\text{VaR} = 24.60 \text{ million}$$

因此，根据计算结果，最接近的选项是 USD 24.00 million。在 99% 的置信水平下，估算的每日 VaR 为 USD 24.00 million。

考察: VaR 的计算。（从一堆数里面找出 VaR）

16. A risk analyst uses historical data from 1,500 days of profit/loss information to calculate a daily VaR at the 95th percentile, which amounts to USD 5 million. Loss observations beyond the 95th percentile are then used to estimate the conditional VaR. If the losses beyond the VaR level, in millions, are USD 6, USD 7, USD 10, USD 13, USD 15, USD 18, USD 21, USD 25, and USD 35, what is the conditional VaR?

- A. USD 10 million
- B. USD 20 million
- C. USD 15 million
- D. USD 17 million

答案:C

解析:

步骤 1：确定 VaR 水平。

- 已知的 VaR 为 USD 5 million，表示在 95% 的置信水平下，损失不会超过这个值。

步骤 2：收集超出 VaR 的损失数据。

- 超出 VaR 水平的损失数据为：USD 6, USD 7, USD 10, USD 13, USD 15, USD 18, USD 21, USD 25, 和 USD 35。

步骤 3：计算条件 VaR。

- 条件 VaR 是超出 VaR 水平的损失的平均值。我们需要计算这些损失的总和，然后除以损失的数量。
 - 计算步骤如下：
$$\text{总损失} = 6 + 7 + 10 + 13 + 15 + 18 + 21 + 25 + 35 = 125$$
 - 计算损失的数量：
$$\text{损失数量} = 9$$
 - 计算条件 VaR：
$$\text{条件 VaR} = \frac{\text{总损失}}{\text{损失数量}} = \frac{125}{9} \approx 13.89 \text{ million}$$

步骤 4：选择最接近的选项。

- 根据计算结果，条件 VaR 约为 USD 13.89 million，最接近的选项是 USD 15 million。

因此，条件 VaR 为 USD 15 million。

考察：VaR 的计算。（从一堆数里面找出 VaR）

17. A risk manager at an investment firm is analyzing the potential losses of a portfolio over different time horizons. The manager has asked a quantitative analyst to calculate the Value at Risk (VaR) for 5-, 10-, 15-, and 20-day periods, assuming that daily returns are independently and identically normally distributed. The manager notices an inconsistency in the analyst’s calculations. Which one of the following VaRs for this portfolio is inconsistent with the others?

- A. $\text{VaR}(5\text{-day}) = \text{USD } 200.00\text{M}$
- B. $\text{VaR}(10\text{-day}) = \text{USD } 316.23\text{M}$
- C. $\text{VaR}(15\text{-day}) = \text{USD } 387.30\text{M}$
- D. $\text{VaR}(20\text{-day}) = \text{USD } 447.21\text{M}$

答案:A

解析：

A 选项正确。

- $\text{VaR}(5\text{-day}) = 200 \Rightarrow$

$$\text{VaR}(1\text{-day}) = \frac{200}{\sqrt{5}} \approx 89.44$$

B 选项错误。

- $\text{VaR}(10\text{-day}) = 316.23 \Rightarrow$

$$\text{VaR}(1\text{-day}) = \frac{316.23}{\sqrt{10}} \approx 100$$

C 选项错误。

- $\text{VaR}(15\text{-day}) = 387.30 \Rightarrow$

$$\text{VaR}(1\text{-day}) = \frac{387.30}{\sqrt{15}} \approx 100$$

D 选项错误。

- $\text{VaR}(20\text{-day}) = 447.21 \Rightarrow$

$$\text{VaR}(1\text{-day}) = \frac{447.21}{\sqrt{20}} \approx 100$$

考察：VaR 的计算。（需要转换到 VaR(1-day)）

18. A risk manager calculates that the VaR of a trading portfolio at 95% confidence level is 12.8 million dollars. Assuming returns follow a one-tailed normal distribution, what would be the closest value of VaR if the confidence level is increased to 99%?

- A. 8.5 million dollars

- B. 14.2 million dollars
- C. 18.1 million dollars
- D. 18.9 million dollars

答案:C

解析:

步骤 1：理解正态分布的标准分位数。

- 95% 置信水平对应的标准正态分位数 $\alpha_{95\%} = 1.645$
- 99% 置信水平对应的标准正态分位数 $\alpha_{99\%} = 2.326$

步骤 2：理解 VaR 的比例关系。

- VaR 与置信水平的标准正态分位数成正比

$$\frac{VaR_{99\%}}{VaR_{95\%}} = \frac{\alpha_{99\%}}{\alpha_{95\%}}$$

步骤 3：计算 99% 置信水平下的 VaR。

- 代入数值计算：

$$VaR_{99\%} = VaR_{95\%} \times \frac{2.326}{1.645}$$
$$VaR_{99\%} = 12.8 \times \frac{2.326}{1.645} = 18.1 \text{ million dollars}$$

考察: VaR 在不同置信水平下的转换计算（重点是正态分布分位数的应用）

19. Sarah, the risk analyst at Quantum Investment Fund, is comparing the risk profiles of two cryptocurrency investment strategies. She collected the following information about the loss distribution of these strategies. Which risk measure would be most appropriate to differentiate their risk severity?

Loss distribution for Strategy A	
Probability	Loss Amount
98.5%	\$15,000
1.5%	\$45,000

Loss distribution for Strategy B	
Probability	Loss Amount
99%	\$15,000
0.5%	\$250,000
0.5%	\$350,000

- A. 98% VaR
- B. 99% VaR
- C. 98% ES
- D. 99% ES

答案:A

解析:

步骤 1：首先计算 Strategy A 的风险指标。

- 98% VaR = \$15,000
- 99% VaR = \$45,000
- 98% ES = $\frac{98.5}{98} \times \$15,000 + \frac{1.5}{98} \times \$45,000 = \$15,459$

- 99% ES = \$45,000

步骤 2：然后计算 Strategy B 的风险指标。

- 98% VaR = \$15,000
- 99% VaR = \$250,000
- $98\% \text{ ES} = \frac{99}{98} \times \$15,000 + \frac{0.5}{98} \times \$250,000 + \frac{0.5}{98} \times \$350,000 = \$18,163$
- $99\% \text{ ES} = \frac{1}{2} \times \$250,000 + \frac{1}{2} \times \$350,000 = \$300,000$

步骤 3：汇总比较。

指标	Strategy A	Strategy B
98% VaR	\$15,000	\$15,000
99% VaR	\$45,000	\$250,000
98% ES	\$15,459	\$18,163
99% ES	\$45,000	\$300,000

步骤 4：结论分析。

- 98% VaR 能够最好地区分两个策略的风险，因为它显示了两个策略在正常市场条件下的风险水平是相同的。
- 其他指标（99% VaR、98% ES、99% ES）都显示 Strategy B 的风险明显高于 Strategy A，这可能导致过度保守的风险评估。

考察: 风险度量指标的选择与计算（VaR 和 ES 的比较分析）

20. Sarah, a quantitative analyst at XYZ Asset Management, is analyzing a structured product with returns following a uniform distribution between -25% and 40%. To assess the risk profile, she needs to calculate both VaR and ES at the 95% confidence level. What are the 95% VaR and 95% ES for this product?

- A. 20.5% and 22.75%
- B. 21.75% and 23.25%
- C. 22.75% and 21.5%
- D. 21.5% and 22.75%

答案:D

解析:

步骤 1：理解均匀分布的特征

- 收益率范围：[-25%, 40%]
- 分布长度：65% (从-25% 到 40%)
- 均匀分布意味着每个点的概率密度相等

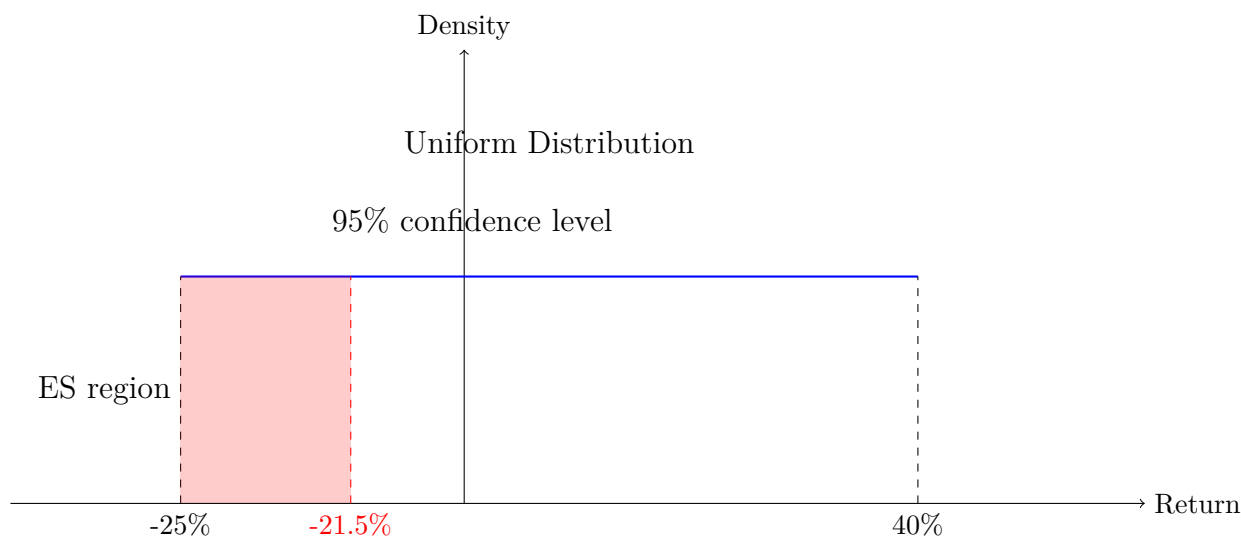
步骤 2：计算 95% VaR

- 使用均匀分布的分位数公式： $\frac{x - (-25\%)}{40\% - (-25\%)} = 1 - 95\% = 5\%$ ，得到 $x = -21.75\%$
- 因此，95% VaR = 21.5%

步骤 3：计算 95% ES

- ES 是 VaR 以外区域的平均损失，在均匀分布中，这等于 VaR 点和最小值的平均值。
- 因此， $ES = \frac{-25\% + (-21.5\%)}{2} = -22.75\%$

步骤 4：图形说明



从图中可以看出：

- 蓝线表示均匀分布的概率密度函数
- 红色虚线表示 95% VaR 的位置 (-21.5%)
- 红色阴影区域表示 ES 计算区间 (从-25% 到-21.5%)
- 由于是均匀分布，ES 就是阴影区域的平均值 (22.75%)

因此，95% VaR = 21.5%，95% ES = 22.75%，正确答案为 D 选项。

考察：均匀分布下 VaR 和 ES 的计算方法。（图形化理解风险度量）

21. In the context of financial risk management, Value at Risk (VaR) is a widely used risk measurement method. A risk management team at a leading financial institution is reviewing the properties and limitations of VaR to ensure its effective application in their investment strategies. During a team meeting, they discuss several statements about VaR. Which of the following statements about VaR is CORRECT?

- A. VaR exhibits strong portfolio diversification effects, consistently reflecting the risk reduction benefits when combining multiple investments.
- B. VaR provides reliable risk estimates across all market conditions due to its robust statistical foundation in normal distribution theory.
- C. The standardized confidence levels and holding periods in VaR calculations must be clearly specified and consistently applied to ensure meaningful risk comparisons across portfolios.
- D. VaR provides an estimate of the actual amount or size of loss an investor might face, reflecting the corresponding risk exposure.

答案:C

解析：

A 选项错误。

- VaR 违反次可加性原则，这意味着当组合多个投资时，组合的 VaR 可能大于各个投资 VaR 的总和。这种情况可能导致对整体风险的高估，因为没有考虑到投资组合的多样化效应。

B 选项错误。

- VaR 在非正态分布条件下可能不可靠，因为它通常假设收益分布是正态的。在极端事件发生时，VaR 可能无法准确捕捉尾部风险，这使得它在处理非正态分布时存在局限性。

C 选项正确。

- VaR 的计算确实需要明确指定和一致应用标准化的置信水平和持有期，这是进行有意义的风险比较的基础。例如，不同投资组合之间的 VaR 比较只有在使用相同的置信水平（如 95% 或 99%）和持有期（如 1 天或 10 天）时才有意义。

- 这种标准化的要求确保了风险度量的可比性和一致性，是 VaR 作为风险管理工具的重要特征。

D 选项错误。

- VaR 并不提供投资者实际损失的金额或大小，它只提供了在给定置信水平下可能的最大损失。相同的 VaR 值并不意味着风险敞口相同，因为不同的回报分布可能具有相同的 VaR，但风险敞口可能非常不同。

考察: VaR 的性质与局限性。（VaR 并不提供投资者实际损失的金额或大小）

22. In the context of financial risk management, understanding the relationship between asset value changes and underlying variables is crucial. A risk management team at a leading financial institution is reviewing the characteristics of linear and nonlinear investment products. During a team meeting, they discuss several statements about these products. Which of the following statements is INCORRECT?

- A. The value change of a linear investment portfolio has a linear relationship with the change in its underlying variables. Futures and forwards are typical examples of linear products.
- B. The value change of a nonlinear investment portfolio does not have a linear relationship with the change in its underlying variables. Options are common examples of nonlinear products.
- C. The first derivative of a linear investment portfolio is constant, while the first derivative of a nonlinear investment portfolio is variable.
- D. A linear investment portfolio exhibits the same risk sensitivity as a nonlinear investment portfolio during market fluctuations.

答案:D

解析:

A 选项正确。

- 线性投资组合的价值变化与其基础变量的价值变化呈线性关系，期货和远期是线性衍生品的典型例子。

B 选项正确。

- 非线性投资组合的价值变化与基础变量的价值变化没有线性关系，期权是常见的非线性衍生品。

C 选项正确。

- 线性投资组合的一阶导数为常数，非线性投资组合的一阶导数为非常数。

D 选项错误。

- 线性投资组合在市场波动时不会表现出与非线性投资组合相同的风险敏感度。线性投资组合的风险敏感度通常是可预测的，而非线性投资组合的风险敏感度可能因市场条件的变化而变化，因此它们的风险敏感度不同。

考察: 线性和非线性投资产品的特性与关系。（非线性投资产品的风险敏感度可能因市场条件的变化而变化）

23. A risk manager is monitoring a domestic equity portfolio valued at USD 100 million using historical VaR with a 100-day lookback period. The initial analysis showed the following extreme negative returns:

Order	Return	Days Ago
1	-10.0%	95
2	-6.3%	17
3	-4.7%	65
4	-4.0%	4
5	-3.8%	5
6	-3.6%	30

Over the next 10 days, four new extreme declines occurred: -25%, -9.5%, -7.8%, and -4.1%. The other six days showed positive returns. Calculate the updated 1-day 95% VaR using the historical approach with equal weights.

- A. USD 3.28 million
- B. USD 4.70 million
- C. USD 10.00 million
- D. USD 25.00 million

答案:B

解析:

步骤 1：整理更新后的极端损失

- 按损失从大到小排序：

$$-25.0\%, -9.5\%, -7.8\%, -6.3\%, -4.7\%, -4.1\%$$

步骤 2：确定 VaR 位置

- 总样本量 = 100 天

$$95\% \text{ 分位数位置} = 100 \times 5\% = 5$$

- 第 5 个最大损失为 4.7%

步骤 3：计算 VaR 金额

$$\text{VaR}(95\%) = 4.7\% \times \text{USD } 100 \text{ million} = \text{USD } 4.70 \text{ million}$$

考察: 历史 VaR 的更新计算（重点是分位数的确定）

2 考点 2:Calculating and Applying VaR

1. A risk analyst at Bergman International Bank is tasked with explaining how to calculate Value at Risk (VaR) for linear derivatives to new junior analysts. Which of the following statements accurately describes the process for calculating VaR for a linear derivative based on the S&P 500 Index?

- A. For a futures contract, the VaR of the S&P 500 Index should be multiplied by a sensitivity factor that indicates the percentage change in the futures contract’s value for a 1% change in the index.
- B. For an options contract, the VaR of the S&P 500 Index should be multiplied by a sensitivity factor that indicates the percentage change in the futures contract’s value for a 1% change in the index.
- C. For a futures contract, the VaR of the S&P 500 Index should be divided by a sensitivity factor that reflects the absolute change in the futures contract’s value for an absolute change in the index.
- D. For an options contract, the VaR of the S&P 500 Index should be divided by a sensitivity factor that indicates the percentage change in the futures contract’s value for a 1% change in the index.

答案:A

解析:

A 选项正确。

- 对于期货合约，计算 VaR 时需要将 S&P 500 指数的 VaR 乘以一个敏感性因子 ($|\Delta|$)，该因子反映了在指数价值变化 1% 时期货合约价值的变化百分比。这是因为期货合约与基础资产之间的关系是线性的，因此可以通过这种方式调整 VaR。

B 选项错误。

- 对于期权合约，VaR 的计算不应使用期货合约的敏感性因子。期权的价值变化是非线性的，因此需要使用期权的德尔塔 (Δ) 来进行调整，而不是期货合约的敏感性因子。

C 选项错误。

- 对于期货合约，VaR 的计算不应通过除以敏感性因子来进行。相反，应该乘以敏感性因子来反映期货合约的价值变化。

D 选项错误。

- 对于期权合约，VaR 的计算同样不应通过除以敏感性因子来进行。期权的价值变化是非线性的，应该使用期权的德尔塔进行调整，而不是期货合约的敏感性因子。

考察: 期货合约和期权合约的 VaR 计算。（区分期权和期货的敏感因子）

2. A risk manager would like to measure VaR for a bond. He notices that the bond has a puttable feature. What effect on the VaR will this puttable feature have?

- A. The VaR will increase.
- B. The VaR will decrease.
- C. The VaR will remain the same.
- D. The effect on the VaR will depend on the volatility of the bond.

答案:B

解析:

B 选项正确。

在这道题中，风险经理注意到债券具有可赎回（puttable）特性。可赎回特性意味着投资者可以在特定条件下将债券“卖回”给发行者。这种特性对 VaR（风险价值）的影响如下：

- 可赎回特性意味着投资者拥有一个选择权
 - 当债券的市场价格下跌时，投资者可以选择将债券赎回，避免进一步的损失。这就像是投资者拥有一个“保险”，可以在不利情况下保护自己的投资。
- 正的伽马效应:
 - 由于可赎回特性，债券的价格对市场波动的反应会变得更加敏感。这种敏感性被称为“伽马”（ Γ ）。伽马是期权定价模型中的一个重要参数，表示期权的德尔塔（ Δ ）对基础资产价格变化的敏感度。
 - 数学上，伽马可以表示为：

$$\Gamma = \frac{\partial^2 C}{\partial S^2}$$

其中， C 是期权的价格， S 是基础资产的价格。正的伽马意味着：

$$\Gamma > 0 \implies \Delta \text{ increases as } S \text{ increases}$$

- 这意味着当市场价格波动时，债券的价值变化会有更好的保护。具体来说，当市场价格下跌时，投资者可以选择将债券赎回，从而避免更大的损失。这种保护机制降低了潜在的损失，导致 VaR（风险价值）降低。
- VaR 的降低:
 - 由于投资者可以在不利情况下将债券赎回，这种保护机制会导致 VaR 降低。换句话说，投资者面临的潜在损失减少了，因此 VaR 也会相应减少。

考察: VaR 的计算。（需要知道 delta 的作用）

3. Two investment banks, Bank X and Bank Y, are calculating the 1-day 99% VaR for a call option on a non-dividend-paying stock. The relevant information is as follows:

- Current stock price: USD 100
- Estimated annual volatility: 20%
- Current option price: USD 4.50
- Option delta: 0.5

Bank X uses a linear approximation method, while Bank Y employs a Monte Carlo simulation for a more detailed analysis. Which bank is likely to report a higher 1-day 99% VaR?

- A. Bank X will report a higher VaR.
- B. Bank Y will report a higher VaR.
- C. Both banks will report the same VaR.
- D. Insufficient information to determine.

答案:B

解析:

B 选项正确。

- 理解计算方法：
 - Bank X 使用线性近似法，这种方法通常会低估风险，因为它假设收益是线性的，未能充分考虑潜在的极端情况。
 - Bank Y 使用蒙特卡洛模拟法，这种方法通过模拟多种市场情景，能够更全面地评估风险，通常会提供更高的 VaR 估计。
- 比较两种方法：
 - 线性近似法在波动性较高的情况下可能会低估 VaR，尤其是在市场波动较大的情况下。
 - 蒙特卡洛模拟法能够考虑到更复杂的市场动态，因此在大多数情况下会给出更高的 VaR 估计。

根据上述分析，Bank Y 更有可能估算出更高的 1 天 99% VaR，因为其方法更为全面和准确。

考察: VaR 的计算。(VaR 和蒙特卡洛模拟的对比)

4. A risk analyst at an investment firm is conducting a historical simulation to estimate the Expected Shortfall (ES) of a portfolio. The portfolio’s value at market close depends on the stock price and the interest rate. The analyst uses data from the most recent 501 trading days, with Day 0 as the first data point and Day 500 as the last. What stock price and interest rate should the analyst use for Day 501?

Day	Stock Price (HKD)	Interest Rate (%)
0	80.00	3.00
1	78.00	3.10
...	...	...
500	100.00	4.00

- A. A. Stock price: HKD 95.00, Interest rate: 4.05%
- B. B. Stock price: HKD 95.00, Interest rate: 4.10%
- C. C. Stock price: HKD 98.00, Interest rate: 4.05%
- D. D. Stock price: HKD 98.00, Interest rate: 4.10%

答案:B

解析:

步骤 1: 确定历史变化率

- 计算从 Day 0 到 Day 1 的股票价格变化率：
$$\text{变化率} = \frac{78 - 80}{80} = -0.025 \quad \text{或} \quad -2.5\%$$
- 计算从 Day 0 到 Day 1 的利率变化：
$$\text{变化率} = 3.10\% - 3.00\% = 0.10\%$$

步骤 2：应用变化率

- 将变化率应用于 Day 500 的股票价格和利率：
- 股票价格的计算：

$$\text{新股票价格} = 100 \times (1 - 0.025) = 97.50 \quad (\text{约为 HKD } 97.50)$$

- 利率的计算：

$$\text{新利率} = 4.00\% + 0.10\% = 4.10\%$$

考察: 历史模拟法。（根据历史数据计算股票价格和利率的变化）

5. A risk management team is discussing different approaches to value at risk (VaR) estimation. Which of the following statements about VaR estimation methods is CORRECT?

- A. Parametric VaR approaches are primarily forward-looking and rely minimally on historical return data for parameter estimation.
- B. GARCH(1,1) models can generate accurate VaR estimates without requiring any historical volatility or return information.
- C. Hybrid approaches combining parametric and nonparametric methods are completely independent of historical market data.
- D. Implied volatility-based VaR estimation is the least dependent on historical returns as it primarily uses current option market prices.

答案:D

解析:

A 选项错误。

- 参数法 VaR 高度依赖历史数据，因为需要使用历史收益率来估计均值、波动率等参数。这些参数的准确性直接影响 VaR 的计算结果。

B 选项错误。

- GARCH(1,1) 模型本质上是一个基于历史数据的模型，需要历史收益率序列来估计波动率持续性参数，其预测能力取决于历史数据的质量。

C 选项错误。

- 混合方法结合了参数法和非参数法的特点，其中非参数部分直接使用历史数据，参数部分也需要历史数据来估计参数，因此无法完全独立于历史市场数据。

D 选项正确。

- 隐含波动率方法主要使用当前期权市场价格，反映了市场对未来波动率的预期，相比其他方法，对历史收益率数据的依赖最小。这种方法更多地体现了市场的前瞻性预期。

考察: VaR 估计方法比较（重点是各种方法对历史数据的依赖程度）

6. A bank’s operational risk department has developed a model for a specific business process. For the next year, the model predicts a 95% probability of no loss occurrence and a 5% probability of experiencing a loss event. If a loss occurs, there are three possible outcomes: a \$12 million loss with 20% probability, a \$20 million loss with 50% probability, and a \$28 million loss with 30% probability. What is the 90% Expected Shortfall (ES) for this operational risk model?

- A. \$10.4 million
- B. \$12.0 million
- C. \$15.6 million
- D. \$20.0 million

答案:A

解析:

步骤 1：理解 90%ES 的计算范围。

$$10\% \text{ 尾部} = \text{无损失区间 (90\%-95\%, 占 5\%)} + \text{损失事件区间 (95\%-100\%, 占 5\%)}$$

步骤 2：计算损失事件的条件期望值。

$$\begin{aligned} E(\text{损失}|\text{损失发生}) &= \sum_{i=1}^3 p_i \times L_i \\ &= 20\% \times \$12M + 50\% \times \$20M + 30\% \times \$28M \\ &= \$2.4M + \$10.0M + \$8.4M = \$20.8M \end{aligned}$$

步骤 3：计算 90% ES。

$$\begin{aligned} ES_{90\%} &= \frac{\text{无损失部分} + \text{损失部分}}{10\%} \\ &= \frac{5\% \times \$0 + 5\% \times \$20.8M}{10\%} \\ &= \frac{\$0 + \$1.04M}{10\%} = \$10.4M \end{aligned}$$

考察: 操作风险的 ES 计算（重点是条件期望和分布组合）

7. A portfolio manager is reviewing the effectiveness of diversification strategies during the 2008 financial crisis. The analysis focuses on how correlations between different asset classes changed during this period and its implications for risk management. Which of the following statements MOST accurately describes the behavior of correlations and risk metrics during financial crises?

- A. Historical correlations between asset classes remain stable during crisis periods, making traditional diversification strategies consistently effective.
- B. Asset correlations tend to increase significantly during market stress, reducing diversification benefits and leading to higher portfolio risk levels.
- C. Market volatilities increase during crises but correlations typically decrease, creating new diversification opportunities.
- D. Standard VaR models automatically adjust for changing correlations during crisis periods, making additional risk adjustments unnecessary.

答案:B

解析:

A 选项错误。

- 这与经验证据相反。研究表明，金融危机期间资产间的相关性会发生显著变化，传统的多元化策略效果会减弱。

B 选项正确。

- 这反映了”相关性趋同”(correlation convergence) 现象：危机期间，大多数资产类别之间的相关性会显著上升；这种相关性增加降低了投资组合多元化的效果，导致在最需要分散化保护时，其效果反而最差。

C 选项错误。

- 这与实际观察相反。危机期间，不仅波动率上升，相关性也会上升，这实际上减少了而不是创造了多元化机会。

D 选项错误。

- 标准 VaR 模型通常使用历史数据估计相关性，无法及时反映危机期间的相关性突变，这正是为什么需要补充使用压力测试等其他风险管理工具。

考察: 金融危机期间的相关性变化（重点是相关性趋同现象及其对风险管理的影响）

8. A risk analyst is evaluating the risk metrics for a trading portfolio using historical simulation. The following 25 ordered percentage returns have been generated:
-18, -15, -12, -10, -8, -6, -5, -5, -3, -2, -1, 0, 0, 1, 2, 3, 5, 7, 8, 10, 12, 15, 17, 20, 22
Required: (1) Calculate the Value at Risk (VaR) at 90% confidence level (2) Calculate the Expected Shortfall (ES) at 90% confidence level

- A. $VaR(90\%) = 8\%$, $ES = 11\%$
- B. $VaR(90\%) = 12\%$, $ES = 15\%$
- C. $VaR(90\%) = 12\%$, $ES = 13.5\%$
- D. $VaR(90\%) = 15\%$, $ES = 18\%$

答案:C

解析:

步骤 1: 计算 90% VaR。

- 对于 $n = 25$ 个观察值，确定分位数位置：

$$\text{位置} = n \times (1 - 90\%) = 25 \times 10\% = 2.5$$

- 取第 3 个最低值：-12%

$$VaR(90\%) = |-12\%| = 12\%$$

步骤 2: 计算 90% ES。

- 计算 VaR 以外损失的平均值：

$$ES(90\%) = \frac{|-18\%| + |-15\%|}{2} = \frac{33\%}{2} = 13.5\%$$

结论: $VaR(90\%) = 12\%$, $ES(90\%) = 13.5\%$

考察: VaR 和 ES 的历史模拟计算（重点是分位数确定和条件期望计算）

9. A risk manager at a global fund believes that the market is about to transition from a bull to a bear market. She needs to select appropriate risk measures and calculation methods to assess the portfolio’s risk exposure during this transition period. Which of the following statements is CORRECT regarding the choice and implementation of risk measures?

- A. VaR and Expected Shortfall are equivalent in measuring tail risk, as both metrics capture the complete distribution of potential losses in the portfolio’s tail.
- B. Historical simulation remains a reliable method for VaR calculation during market regime changes, as past price patterns and correlations will continue to be representative.
- C. Monte Carlo simulation with appropriate distributional assumptions and volatility adjustments is suitable for VaR calculation during market transitions, as it allows for flexible modeling of changing market conditions.
- D. For portfolios containing options, the delta-normal approach without gamma adjustment provides accurate VaR estimates, as option risks are primarily linear in nature.

答案:C

解析:

A 选项错误。

- VaR 只衡量特定分位点的损失，而 ES 才能完整描述尾部风险分布。两者在尾部风险刻画能力上存在本质差异。

B 选项错误。

- 市场转型期间，历史数据的代表性显著下降。历史模拟方法过度依赖过去数据，无法准确反映市场机制的根本变化。

C 选项正确。

- 蒙特卡洛模拟允许灵活调整波动率和相关性假设，可以通过参数调整反映市场状态转换，适合建模转型期间的风险特征。

D 选项错误。

- 期权具有显著的非线性特征，忽略 gamma 效应会严重低估风险。在市场剧烈波动时期，这种低估会更加严重。

考察：风险度量方法选择（重点是不同方法在市场转型期的适用性）

10. A portfolio manager at a university endowment fund is comparing different VaR calculation methods for their €80,000,000 equity portfolio. The portfolio has the following characteristics:

- Daily expected return: 0.05%
- Daily volatility: 0.85%
- Significance level: 2% (z-value = 2.05)

The manager also collected 250 daily historical returns. The six lowest returns are:

Rank	Return
1	-3.25%
2	-2.88%
3	-2.65%
4	-2.42%
5	-2.28%
6	-2.15%

Compare the daily VaR estimates using both delta-normal and historical simulation methods.

- A. Both methods give the same VaR estimate of €2.12 million
- B. Delta-normal: €1.68 million; Historical: €2.12 million
- C. Delta-normal: €1.32 million; Historical: €1.82 million
- D. Delta-normal: €2.12 million; Historical: €1.68 million

答案:B

解析：

步骤 1：计算 Delta-normal VaR

$$\begin{aligned} \text{VaR}_{\text{Delta-normal}} &= [-(\mu - z\sigma)] \times \text{Portfolio Value} \\ &= [-(0.05\% - 2.05 \times 0.85\%)] \times €80,000,000 \\ &= [0.05\% + 1.74\%] \times €80,000,000 = €1.68 \text{ million} \end{aligned}$$

步骤 2：计算 Historical VaR

$$\begin{aligned} \text{观察数} &= 250, \text{分位点} = 250 \times 2\% = 5 \\ \text{第 5 个最低收益率} &= -2.65\% \end{aligned}$$

$$\text{VaR}_{\text{Historical}} = 2.65\% \times €80,000,000 = €2.12 \text{ million}$$

考察：VaR 计算方法比较（参数法 vs 历史模拟法）

11. A junior risk analyst at an asset management firm is currently evaluating different methods for assessing portfolio risk. The analyst is particularly interested in historical simulation due to its straightforward approach and reliance on past data. During a team meeting, the analyst presents several statements about historical simulation to discuss its advantages and limitations. Which of the following statements about historical simulation is correct?

- A. Historical simulation provides dynamic updates to volatility and correlation estimates through continuous market feedback mechanisms.
- B. Historical simulation incorporates time-varying weights to reflect the relative importance of recent versus distant observations.
- C. Historical simulation does not require any model assumptions.
- D. Historical simulation generates reliable future market predictions through comprehensive pattern recognition algorithms.

答案:C

解析:

A 选项错误。

- 历史模拟法是一种静态方法，它不能动态更新波动率和相关性估计。它仅仅使用历史数据进行模拟，无法通过市场反馈机制来更新估计值。

B 选项错误。

- 标准历史模拟法对所有观察值赋予相同的权重，并不会根据观察值的时间远近来调整权重。这种平等权重的特性可能会导致无法充分反映最近市场变化的影响。

C 选项正确。

- 历史模拟法作为一种非参数方法，确实不需要依赖任何特定的模型假设。它直接使用历史数据来生成可能的市场情景，这是其主要优势之一。

D 选项错误。

- 历史模拟法并不包含模式识别算法，它仅仅基于历史数据进行简单的重采样。它不能保证对未来市场行为做出可靠预测，因为历史模式可能不会在未来重复。

考察: 历史模拟法的特征（作为一种非参数方法的优势）

12. A junior risk analyst at an investment firm is evaluating the delta-normal method for risk assessment. The analyst presents several statements about the delta-normal method during a team meeting. Which of the following statements is CORRECT?

- A. The delta-normal method performs well for linear portfolios and can be applied to nonlinear portfolios.
- B. The delta-normal method primarily relies on historical volatility patterns to determine the distribution of portfolio returns.
- C. In the delta-normal method, actual changes in all risk factors are more practical than percentage changes.
- D. The delta-normal method achieves optimal results through dynamic correlation adjustments in varying market conditions.

答案:A

解析:

A 选项正确。

- Delta-normal 方法确实适用于线性依赖于市场变量的投资组合，并且在处理这类投资组合时表现良好。此外，虽然它主要用于线性投资组合，但也可以通过近似方法用于非线性投资组合。（教材的观点是可以用在非线性投资组合上，但是只是个近似值，倒不是说不能用。）

B 选项错误。

- Delta-normal 方法主要基于正态分布假设，而不是依赖历史波动率模式。它假设投资组合价值的变化服从正态分布，这是一个前提假设，而不是通过历史数据得出的结论。

C 选项错误。

- Delta-normal 方法并不是对所有风险因素都假设实际变化比百分比变化更实用。实际上，风险因素的适用性取决于具体情况：

- 对于股票和商品价格等风险因素，百分比变化可能更实用。
- 对于利率和信用利差等风险因素，实际变化可能更实用。

因此，不能一概而论地说所有风险因素的实际变化都比百分比变化更实用。

D 选项错误。

- Delta-normal 方法使用固定的相关性假设，并不会动态调整相关性。它假设相关性在计算期间保持稳定，这实际上是该方法的一个局限性，而不是优势。

考察: Delta-normal 方法的局限性（它主要用于线性投资组合，但也可以通过近似方法用于非线性投资组合）

13. Consider a portfolio consisting of a \$10,000 investment in asset X and a \$20,000 investment in asset Y. Assume the daily volatilities of X and Y are 1% and 2% respectively, and the correlation coefficient between their returns is 0.3. Which of the following options correctly describes the portfolio's five-day VaR and ES at a 97% confidence level?

- A. VaR is \$1852.4, ES is \$2233.8.
- B. VaR is \$2233.8, ES is \$1852.4.
- C. VaR is \$440.5, ES is \$984.9.
- D. VaR is \$984.9, ES is \$440.5.

答案:A

解析:

理论公式:

$$\begin{aligned}\text{VaR} &= \mu_P - z\sigma_P \\ \text{ES} &= \mu_P + \sigma_P \frac{e^{-(z^2/2)}}{(1-x)\sqrt{2\pi}}\end{aligned}$$

其中， x 是置信水平， z 是正态分布中超过该概率的点。

步骤 1：计算投资组合的日变化标准差

- $\sigma_P = \sqrt{100^2 + 400^2 + 2 \times 100 \times 400 \times 0.3} = 440.5$

步骤 2：计算五天变化的标准差

- $\sigma_{5P} = \sqrt{5} \times 440.5 = 984.9$

步骤 3：计算五天 VaR（97% 置信水平）

- $\text{VaR}_{5P} = 1.88 \times 984.9 = 1852.4$

步骤 4：计算五天 ES（97% 置信水平）

- $\text{ES}_{5P} = 984.9 \frac{e^{-1.88^2/2}}{0.03\sqrt{2\pi}} = 2233.8$

考察: VaR 和 ES 计算（重点是使用 Delta-normal 方法进行计算）

14. A quantitative analyst at a global investment bank is evaluating different VaR calculation methods for their complex derivatives portfolio. During a team discussion about Monte Carlo simulation, which of the following statements about this method is CORRECT?

- A. The Monte Carlo simulation primarily relies on historical data patterns, making it more reliable than parametric approaches for risk factor distribution estimation.
- B. The Monte Carlo simulation's model risk can be effectively eliminated by increasing the number of simulations and using more sophisticated probability distributions.
- C. The Monte Carlo simulation is computationally intensive, making it suitable for scenarios requiring quick results.

D. Using historical data to estimate risk factor characteristics may not accurately predict future market changes.

答案:D

解析:

A 选项错误。

- 蒙特卡洛模拟并不主要依赖历史数据模式，而是通过假设风险因子的概率分布来生成随机样本。虽然历史数据可以用来估计分布参数，但模拟的核心是基于概率分布的随机抽样过程。

B 选项错误。

- 增加模拟次数和使用更复杂的概率分布可以改善模拟结果，但不能完全消除模型风险。模型风险源于对市场行为的基本假设，这种风险无法仅通过技术改进来消除。

C 选项错误。

- 蒙特卡洛模拟法计算量大，适用于需要详细结果的场景，而不是快速结果。由于需要生成大量的随机样本来进行模拟，蒙特卡洛模拟通常需要较长的计算时间，因此不适合需要快速结果的场景。

D 选项正确。

- 使用历史数据估计风险因子的特性，未必能准确预测未来市场变化。历史数据反映的是过去的市场行为，而未来市场可能会受到不同因素的影响，因此仅依赖历史数据可能无法准确预测未来的市场变化。

考察: 蒙特卡洛模拟法的局限性（计算量大，不适合快速结果）

15. A risk manager at a global investment bank is evaluating different VaR calculation methods for their diverse portfolio. During their analysis, they notice that each method has its own strengths and limitations in capturing different types of risks. After reviewing the characteristics of Delta-Normal, Historical Simulation, and Monte Carlo Simulation methods, which of the following statements about these approaches is CORRECT?

- A. The Delta-Normal method provides comprehensive risk assessment for option portfolios through its linear approximation approach, making it particularly suitable for complex derivatives.
- B. The Historical Simulation method maintains consistent computational efficiency across all portfolio types, with enhanced accuracy for structured products.
- C. The Monte Carlo Simulation method is computationally complex but can achieve good VaR precision through multiple iterations.
- D. The Delta-Normal method performs excellently in VaR precision over short windows, while the Historical Simulation method has advantages in handling time-varying risks and unusual events.

答案:C

解析:

A 选项错误。

- Delta-Normal 方法虽然计算简单，但其线性近似方法在处理期权等非线性产品时存在明显局限性。它可能会低估非线性产品的风险，特别是在处理复杂衍生品时的高值概率估计不准确。

B 选项错误。

- 历史模拟法的计算效率实际上会随着投资组合复杂度增加而降低，对于结构化产品尤其如此。此外，其准确性很大程度上依赖于历史数据的代表性。

C 选项正确。

- 蒙特卡洛模拟法确实计算复杂，需要大量的计算资源，但通过增加模拟次数，可以提高 VaR 计算的精度。这种通过迭代提高精度的特性是该方法的重要优势。

D 选项错误。

- Delta-Normal 方法在短窗口期的 VaR 精度并不一定优于其他方法，尤其是在处理非线性风险时。历史模拟法在处理时间变化风险和不寻常事件时确实具有优势，但 Delta-Normal 方法在这方面并不突出。

考察: 历史模拟法、蒙特卡洛模拟法和 Delta-Normal 三种方法的比较（综合起来考察）

16. A junior risk analyst at an asset management firm is currently evaluating different methods for assessing portfolio risk. The analyst is particularly interested in historical simulation due to its straightforward approach and reliance on past data. During a team meeting, the analyst presents several statements about historical simulation to discuss its advantages and limitations. Which of the following statements about historical simulation is CORRECT?

- A. Worst Case Analysis primarily serves as a predictive tool for future market movements, providing reliable forecasts through statistical extrapolation of historical extreme events.
- B. Contagion effects occur when both volatility and correlation increase, enhancing the benefits of diversification. VaR and ES can be used to simulate this effect.
- C. Correlation patterns maintain stability across different market conditions, allowing for consistent risk estimation regardless of market stress levels.
- D. When calculating VaR or ES, we should estimate correlations under extreme market conditions rather than normal market conditions.

答案:D

解析:

A 选项错误。

- 最坏情况分析主要是作为 VaR 的补充措施，用于理解历史极端损失的分布特征，而不是预测未来市场走势。它的主要功能是帮助理解潜在风险暴露，而不是提供可靠的未来预测。

B 选项错误。

- 传染效应通常在波动率和相关性都增加时发生，这会抵消任何分散化的好处，而不是增强分散化的好处。VaR 和 ES 可以用于模拟危机事件中可能发生的传染效应。

C 选项错误。

- 相关性模式在不同市场条件下并不稳定。特别是在压力市场条件下，相关性通常会显著增加，这种现象被称为相关性崩溃，会导致传统的风险估计方法失效。

D 选项正确。

- 在计算 VaR 或 ES 时，我们应该估计极端市场条件下的相关性，而不是正常市场条件下的相关性。这是因为在极端市场条件下，相关性可能会显著变化。

考察: 压力环境的情况（传染效应会抵消分散效应）

3 考点 3:Measuring and Monitoring Volatility

1. A financial analyst is evaluating different models for estimating the volatility of asset returns in order to improve risk management strategies. The analyst comes across four statements regarding these models. Which of the following statements is INCORRECT?

- A. In the exponentially weighted moving average (EWMA) model, some positive weight is assigned to the long-run average variance.
- B. In the EWMA model, the weights assigned to observations decrease exponentially as the observations become older.
- C. In the GARCH (1,1) model, a positive weight is estimated for the long-run average variance.

D. In the GARCH (1,1) model, the weights estimated for observations decrease exponentially as the observations become older.

答案:A

解析:

A 选项错误。

- 在 EWMA 模型中，更新波动率时并不涉及长期平均方差。换句话说，分配给长期平均方差的权重为零。公式为：

$$\sigma_t^2 = (1 - \lambda) \sum_{i=0}^{t-1} \lambda^i r_{t-i}^2$$

其中， λ 是衰减因子，长期平均方差不参与计算。

B 选项正确。

- 在 EWMA 模型中，分配给观察值的权重确实是随着观察值变得更旧而指数递减的。公式为：

$$w_i = \lambda^i$$

这表明较新的观察值对波动率的影响更大。

C 选项正确。

- 在 GARCH (1,1) 模型中，确实为长期平均方差估计了一个正权重。公式为：

$$\sigma_t^2 = \alpha_0 + \alpha_1 r_{t-1}^2 + \beta_1 \sigma_{t-1}^2$$

其中， α_0 是长期平均方差的估计值。

D 选项正确。

- 在 GARCH (1,1) 模型中，观察值的权重确实是随着观察值变得更旧而指数递减的。公式为：

$$w_i = \alpha_1 + \beta_1$$

这表明较新的观察值对波动率的影响更大。

考察: 对波动率估计模型（如 EWMA 和 GARCH）的理解。（理解在长期平均方差和观察值权重分配方面的特性）

2. A junior risk analyst is tasked with modeling the volatility of a specific market variable. The analyst is contemplating the use of either the EWMA model or the GARCH (1,1) model. Which of the following statements accurately reflects the characteristics of these models?

- A. The EWMA model can be viewed as a special case of the GARCH (1,1) model, assuming that the long-run volatility is zero.
- B. The variance estimated from the GARCH (1,1) model is a weighted average of the prior day’s estimated variance and the prior day’s squared return.
- C. The GARCH (1,1) model assigns a higher weight to the prior day’s estimated variance than the EWMA model.
- D. The variance estimated from the EWMA model is a weighted average of the prior day’s estimated variance and the prior day’s observed squared return.

答案:D

解析:

A 选项错误。

- EWMA 模型确实是 GARCH (1,1) 模型的特例，但假设长期波动率为零的说法不准确。公式为：

$$\sigma_t^2 = (1 - \lambda) \sum_{i=0}^{t-1} \lambda^i r_{t-i}^2$$

其中， λ 是衰减因子，长期波动率的权重为零，但其他参数的权重之和为 1。

B 选项错误。

- 从 GARCH (1,1) 模型估计的方差不仅包括前一天的估计方差和前一天的平方收益，还包括长期平均方差的权重。公式为：

$$\sigma_t^2 = \alpha_0 + \alpha_1 r_{t-1}^2 + \beta_1 \sigma_{t-1}^2$$

其中， α_0 是长期平均方差的估计。

C 选项错误。

- GARCH (1,1) 模型对前一天估计方差的权重与 EWMA 模型的权重不能直接比较，因为这取决于具体的参数配置。公式为：

$$w = \alpha_1 + \beta_1$$

这表明权重的比较需要具体的参数值。

D 选项正确。

- 从 EWMA 模型估计的方差确实是前一天估计方差和前一天观察到的平方收益的加权平均。公式为：

$$\sigma_t^2 = (1 - \lambda)r_{t-1}^2 + \lambda\sigma_{t-1}^2$$

这表明 EWMA 模型使用了前一天的平方收益和估计方差。

考察: 对 EWMA 和 GARCH (1,1) 模型的理解。（理解在波动率估计中的特性和相互关系）

3. A risk analyst is using historical simulation with declining weights ($\lambda = 0.97$) to estimate the daily 5% VaR for a portfolio. The following table shows the ordered returns and their weights:

Return (%)	Days ago	Weight	Cumulative weight
-0.93	19	0.01734	0.01734
-0.89	92	0.00188	0.01922
-0.86	82	0.00254	0.02176
-0.80	78	0.00287	0.02463
-0.74	62	A	0.02931
-0.43	98	0.00156	0.03088
-0.31	76	0.00305	0.03393
-0.29	84	0.00239	0.03633
-0.15	59	0.00513	0.04145
-0.07	37	0.01002	0.05147

Which of the following statements is INCORRECT?

- A. The missing weight in cell A can be calculated as $w_{62} = (1 - 0.97) \times 0.97^{62-1} = 0.00468$
- B. Based on the cumulative weights, the 5% VaR lies between -0.07% and -0.15%
- C. Using equal-weighted historical simulation would result in a higher VaR estimate than using declining weights
- D. A smaller λ value leads to a slower decline in the weights as we move further into the past

答案:D

解析:

A 选项正确。

- 权重计算公式： $w_t = (1 - \lambda)\lambda^{t-1}$
- 代入 $\lambda = 0.97$, $t = 62$:

$$w_{62} = (1 - 0.97) \times 0.97^{61} = 0.03 \times 0.97^{61} = 0.00468$$

- $\therefore$ 单个权重计算结果准确

B 选项正确。

- 给定：5%VaR 对应累积权重 ≈ 0.05
- 观察数据： $0.04145 < 0.05 < 0.05147$
- 对应收益率： $-0.15\% > \text{VaR} > -0.07\%$
- $\therefore$ VaR 确实位于 $[-0.15\%, -0.07\%]$ 区间

C 选项正确。

- 等权重法： $\forall t, w_t = \frac{1}{n}$ ，其中 n 为样本数
- 衰减权重法： $w_t = (1 - \lambda)\lambda^{t-1}$ ，且 $w_1 > w_2 > w_3 > \dots$
- 对于远期数据：

$$w_t^{\text{等权}} = \frac{1}{n} > w_t^{\text{衰减}} = (1 - \lambda)\lambda^{t-1}$$

- $\therefore$ 等权重法给予远期极端值更高权重 $\Rightarrow$ VaR 估计更高

D 选项错误。

- 考虑两个 λ 值： $\lambda_1 = 0.8$ 和 $\lambda_2 = 0.97$
- 对于任意时点 t，权重比值：

$$\frac{w_t(\lambda_1)}{w_t(\lambda_2)} = \frac{(1 - 0.8)}{(1 - 0.97)} \times (\frac{0.8}{0.97})^{t-1}$$

- 当 t 增大时，由于 $\frac{0.8}{0.97} < 1$ ：

$$\lim_{t \rightarrow \infty} (\frac{0.8}{0.97})^{t-1} = 0$$

- $\therefore \lambda$ 值越小，权重衰减越快，选项说法错误

考察: 衰减权重 VaR 计算（重点是 λ 值对权重衰减速度的影响）

4. A portfolio manager estimates the following GARCH(1,1) model based on the daily returns of a stock:

$$\sigma_t^2 = 0.00016 + 0.03\mu_{t-1}^2 + 0.85\sigma_{t-1}^2$$

Given the following information:

Annualized return	18%
Risk-free rate	4%
Trading days per year	250 days

Assuming daily volatility is independently identically distributed, using the long-term daily volatility from the GARCH(1,1) model, which of the following is closest to the annualized Sharpe ratio of the stock?

- A. 0.25
- B. 0.45
- C. 0.65
- D. 0.85

答案:B

解析:

步骤 1: 计算长期日波动率

- GARCH(1,1) 模型参数：

$\omega = 0.00016$ （常数项）

- $\alpha = 0.03$ (ARCH 项系数)
- $\beta = 0.85$ (GARCH 项系数)

- 长期波动率计算：

$$\sigma_L = \sqrt{\frac{\omega}{1 - \alpha - \beta}} = \sqrt{\frac{0.00016}{1 - 0.03 - 0.85}} = 3.65\%$$

步骤 2：计算年化夏普比率

- 年化调整： $\sigma_{\text{annual}} = \sigma_L \times \sqrt{250}$
- 夏普比率计算：

$$SR = \frac{18\% - 4\%}{3.65\% \times \sqrt{250}} = 0.45$$

考察：GARCH 模型长期波动率与夏普比率计算（综合考察）

5. Suppose σ_t^2 is the estimated variance at time t and μ_t is the realized return at t. Which of the following GARCH(1,1) models will take the longest time to revert to its long-run volatility mean?

- A. $\sigma_t^2 = 0.04 + 0.02\mu_{t-1}^2 + 0.92\sigma_{t-1}^2$
- B. $\sigma_t^2 = 0.02 + 0.04\mu_{t-1}^2 + 0.94\sigma_{t-1}^2$
- C. $\sigma_t^2 = 0.03 + 0.02\mu_{t-1}^2 + 0.95\sigma_{t-1}^2$
- D. $\sigma_t^2 = 0.03 + 0.03\mu_{t-1}^2 + 0.93\sigma_{t-1}^2$

答案:B

解析：

步骤 1：计算各模型的 Persistence Level

- Persistence Level = $\alpha + \beta$ ，其中：
 - α 是 ARCH 项系数 (μ_{t-1}^2 的系数)
 - β 是 GARCH 项系数 (σ_{t-1}^2 的系数)
- 各选项的 Persistence Level：

选项	α	β	$\alpha + \beta$
A	0.02	0.92	0.94
B	0.04	0.94	0.98
C	0.02	0.95	0.97
D	0.03	0.93	0.96

步骤 2：分析结果

- Persistence Level 越大，波动率回归均值的速度越慢，B 选项的 Persistence Level 最大（0.98），因此 B 选项的模型回归均值的时间最长。

考察：GARCH 模型的持续性水平（Persistence Level）分析

6. A risk analyst is using a GARCH model to estimate volatility for VaR calculations. The current volatility exhibits mean-reverting characteristics. When extending the one-day VaR forecast to a t-day horizon under these conditions, which of the following statements is correct?

- A. The t-day VaR will always be less than $\sqrt{t}$ times the one-day VaR, due to mean reversion.
- B. The t-day VaR will exactly equal $\sqrt{t}$ times the one-day VaR, following the square root of time rule.
- C. The t-day VaR will always be greater than $\sqrt{t}$ times the one-day VaR, due to volatility clustering.

D. The relationship between t-day VaR and one-day VaR depends on whether current volatility is above or below its long-run mean.

答案:D

解析:

均值回归知识点

- 时间延展性分析：

标准情况： $VaR(t) = \sqrt{t} \times VaR(1)$

- 均值回复效应：

$$\text{当前波动率} \begin{cases} > \text{长期均值} \Rightarrow \text{实际 } VaR(t) < \sqrt{t} \times VaR(1) \\ < \text{长期均值} \Rightarrow \text{实际 } VaR(t) > \sqrt{t} \times VaR(1) \end{cases}$$

- 原因说明：

均值回复特性使得波动率趋向于长期均值水平，

因此时间延展效应取决于当前波动率相对于长期均值的位置

A 选项错误。

- 这种说法过于绝对。在均值回复模型中，当当前波动率低于长期均值时，t 日 VaR 实际上可能大于 $\sqrt{t}$ 倍的单日 VaR。这是因为波动率会向上调整至其长期均值水平。

B 选项错误。

- $\sqrt{t}$ 规则仅适用于收益率独立同分布的假设情况。在 GARCH 模型中，由于存在均值回复特性，这个简单的时间延展规则并不适用。波动率的变化会受到其当前水平与长期均值之间差异的影响。

C 选项错误。

- 这种说法同样过于绝对。当当前波动率高于长期均值时，由于均值回复特性，t 日 VaR 实际上可能小于 $\sqrt{t}$ 倍的单日 VaR，因为波动率会向下调整至其长期均值水平。

D 选项正确。

- 这准确反映了 GARCH 模型中均值回复的特性。当前波动率与长期均值的相对位置决定了未来波动率的变化方向：
- 如果当前波动率高于长期均值，未来波动率趋于下降，导致实际 VaR 小于 $\sqrt{t}$ 规则预测值。
- 如果当前波动率低于长期均值，未来波动率趋于上升，导致实际 VaR 大于 $\sqrt{t}$ 规则预测值。

考察: GARCH 模型与 VaR 时间延展（重点是均值回复对 VaR 计算的影响）

7. A risk analyst is using an exponential smoothing (EWMA) approach to estimate daily volatility, with a lambda parameter of 0.920. If today is denoted as time t, and we are considering the weights assigned to past returns, what are the weights for the returns at time (t-1) and (t-2) respectively?

- A. Weight(t-1) = 8.00% and Weight(t-2) = 1.25%
- B. Weight(t-1) = 8.00% and Weight(t-2) = 7.36%
- C. Weight(t-1) = 65.25% and Weight(t-2) = 52.41%
- D. Weight(t-1) = 92.00% and Weight(t-2) = 84.64%

答案:B

解析:

步骤 1：理解 EWMA 权重的基本原理。

- EWMA 的基本公式：

$$\sigma_t^2 = \lambda \sigma_{t-1}^2 + (1 - \lambda) r_{t-1}^2$$

- 权重计算的一般公式：

$$w_i = (1 - \lambda)\lambda^i, \quad i \geq 0$$

其中，i 表示从当前时点往回数的期数

步骤 2：计算 (t-1) 期的权重。

- 代入 $i = 0$ 到权重公式：

$$w_{t-1} = (1 - \lambda)\lambda^0 = 1 - 0.920 = 0.080 = 8.00\%$$

步骤 3：计算 (t-2) 期的权重。

- 代入 $i = 1$ 到权重公式：

$$w_{t-2} = (1 - \lambda)\lambda^1 = 0.080 \times 0.920 = 0.0736 = 7.36\%$$

步骤 4：验证结果的合理性。

- 权重递减规律： $w_i = w_{i-1} \times \lambda$
- 验证： $7.36\% = 8.00\% \times 0.920$
- 这符合 EWMA 对近期数据赋予更高权重的特性

考察：EWMA 模型中权重计算（重点是权重递减规律和计算方法）

8. A junior market risk analyst is studying the mechanics of the EWMA approach for estimating volatility. The analyst observes that the approach applies various weights to a series of historical returns, and the return needed to update the EWMA calculation is the most recent day's squared return. Which of the following statements is correct?

- A. Daily returns prior to the most recent day have no influence on the current variance rate estimate in the EWMA calculation.
- B. Daily returns prior to the most recent day are incorporated into the EWMA calculation through the smoothing parameter (λ).
- C. Daily returns prior to the most recent day are reflected in the EWMA calculation by the most recent day's squared return.
- D. Daily returns prior to the most recent day are reflected in the EWMA calculation by the previous variance rate estimate.

答案:B

解析：

EWMA 公式：

$$\sigma_n^2 = (1 - \lambda)r_{n-1}^2 + \lambda\sigma_{n-1}^2$$

其中， σ_n^2 是当前的方差估计， r_{n-1}^2 是前一天的平方收益， σ_{n-1}^2 是前一日方差估计， λ 是平滑参数。

A 选项错误。

- 历史收益通过平滑参数 λ 影响当前方差估计。具体来说， λ 的值决定了历史收益的权重，较小的 λ 会使得最近的收益对方差估计的影响更大。

B 选项正确。

- λ 决定了历史收益在当前方差估计中的权重。较大的 λ 意味着历史收益的影响较小，而较小的 λ 则意味着最近的收益对方差估计的影响更大。这种加权方式使得 EWMA 方法能够更灵活地适应市场波动。

C 选项错误。

- 最近一天的平方收益 r_{n-1}^2 仅用于更新当前方差估计 σ_n^2 ，而不是直接反映之前的收益，而是作为更新当前方差估计的一个组成部分。EWMA 方法的核心在于如何通过平滑参数 λ 来加权历史收益的影响，而不是单纯依赖最近一天的收益。

D 选项错误。

- 虽然前一方差估计确实包含了之前的收益信息，但它并不是直接反映在 EWMA 计算中的方式。前一方差估计是通过历史收益和 λ 计算得出的，反映了所有历史收益的综合影响，而不是单一的历史收益。

考察: VaR 的计算。(EWMA 的公式理解)

9. A market risk analyst is studying volatility patterns in financial markets and discussing the concept of regime switching. The analyst needs to understand how market volatility changes under different conditions and what triggers these changes. Which of the following statements about regime switching in volatility modeling is INCORRECT?

- A. High volatility periods tend to be followed by high volatility periods in normal market conditions, demonstrating the clustering nature of market volatility.
- B. Regime switching refers to gradual and predictable changes in market volatility that allow investors to adjust their positions systematically.
- C. An unexpected central bank announcement can trigger an immediate regime switch in volatility, causing market turbulence followed by stabilization.
- D. Low volatility periods are typically followed by low volatility periods under normal circumstances, reflecting the persistence of market conditions.

答案:B

解析:

A 选项正确。

- 在正常市场条件下，高波动率期间往往会持续一段时间，这反映了市场波动率的聚集性特征。这种现象在金融市场建模中是一个基本假设。

B 选项错误。

- Regime switching 指的是波动率的快速且突然的改变，而不是渐进和可预测的变化。这种突然的转变往往是由重大事件或意外消息触发的，具有不可预测性。

C 选项正确。

- 央行的意外公告确实可能导致市场波动率的突然转换，这是 regime switching 的典型例子。市场在消化新信息后，波动率可能会逐渐回归正常水平。

D 选项正确。

- 在正常市场环境下，低波动率期间倾向于持续，这反映了市场状态的持续性特征，直到出现触发 regime switching 的重大事件。

考察: 波动率模型中的 Regime Switching 概念 (regime switching 指的是波动率的快速且突然的改变，而不是渐进和可预测的变化)

10. A financial analyst at a hedge fund is monitoring market volatility to adjust the fund’s risk management strategies. The analyst reviews historical data and notices that the average standard deviation of asset returns is 1% per day. However, the current volatility monitoring program indicates that the volatility has increased to 2%. Based on the following market volatility monitoring results, which statement is correct?

- A. In an unconditionally normal model, the probability distribution of daily returns assumes a standard deviation of 1%.
- B. The conditionally normal model maintains stable volatility patterns through market cycles, providing consistent risk estimates.
- C. The current market volatility of 2% represents a temporary deviation that should be disregarded in favor of the long-term average of 1%.
- D. If the conditional distribution is normal, returns have similar means and similar volatility.

答案:A

解析:

A 选项正确。

- 无条件正态模型假设每天的收益概率分布具有相同的标准差，这里为 1%。这种模型不考虑市场条件的变化。

B 选项错误。

- 条件正态模型的特点恰恰是允许波动率随市场条件变化而变化，而不是保持稳定。该模型的优势在于能够捕捉市场波动性的动态变化，而不是提供一致的风险估计。

C 选项错误。

- 当前 2% 的市场波动率是重要的市场信号，不应被忽视。在风险管理中，应该根据当前市场条件调整风险评估，而不是简单地依赖长期平均值。

D 选项错误。

- 即使条件分布是正态分布，收益的均值和波动性也可能不同，特别是在市场条件变化时。条件分布允许均值和波动性随市场状况变化。

考察: conditionally normal model 和 unconditionally normal model（学会比较分析有条件分布带来的影响）

11. A junior risk analyst at an asset management firm is currently evaluating different methods for assessing portfolio risk. The analyst is particularly interested in understanding the differences between implied volatility and historical volatility to enhance the firm's risk assessment models. During a team meeting, the analyst presents several statements about these volatilities. Which of the following statements is CORRECT?

- A. Historical volatility, calculated from past data, is forward-looking, while implied volatility is retrospective.
- B. Implied volatility of options maintains consistent values across different maturities due to the market's uniform expectations of future volatility.
- C. Implied volatility provides a better estimate of actual future volatility than historical volatility.
- D. The conversion between annual and daily implied volatility requires adjustments based on market liquidity conditions rather than trading days.

答案:C

解析:

A 选项错误。

- 历史波动率是基于过去的市场数据计算得出的，反映了过去一段时间内资产价格的波动情况，因此是回顾性的。
- 隐含波动率是从期权市场价格中推导出来的，反映了市场参与者对未来波动性的预期，因此是前瞻性的。

B 选项错误。

- 隐含波动率在不同到期日之间通常会有差异，这反映了市场对不同时间段的波动性预期不同。市场预期并非统一，而是会根据期限结构呈现出不同的波动率水平。

C 选项正确。

- 隐含波动率通常被认为比历史波动率提供了更好的实际波动率估计，因为它反映了市场对未来波动性的预期。

D 选项错误。

- 隐含波动率的年化与日度转换主要基于交易日数（通常为 252 天的平方根），而不是市场流动性条件。这是一个标准的时间调整方法，与市场流动性无关。

考察: 历史波动率和隐含波动率的区别（历史波动率是回顾性，隐含波动率是前瞻性。）

12. A quantitative analyst at a financial institution is exploring the use of Multivariate Density Estimation (MDE) for portfolio analysis. The analyst presents several statements about MDE during a team discussion. Which of the following statements is INCORRECT?

- A. Multivariate Density Estimation adjusts weights by comparing historical returns with current market factors.
- B. Historical returns with high factor similarity to current market conditions are given greater weight.
- C. Multivariate Density Estimation considers the time distance of historical data.
- D. Multivariate Density Estimation is applicable to factors such as interest rate levels and GDP growth rates.

答案:C

解析:

A 选项正确。

- 多元密度估计法通过比较历史回报和当前市场的因子来调整权重。这种方法确保了分析结果与当前市场状况的相关性。

B 选项正确。

- 当前市场情况的因子相似性高的历史回报会被赋予更大的权重。这符合 MDE 方法的描述，强调因子相似性的重要性。

C 选项不正确。

- 多元密度估计法不考虑历史数据的时间距离，而是关注历史数据与当前市场条件的相似性。因此，时间距离不是 MDE 方法的核心关注点。

D 选项正确。

- 多元密度估计法适用于利率水平和 GDP 增长率作为影响因子。这表明它可以处理多种影响因子，适用于复杂的市场分析。

考察: MDE 方法 (时间距离不是 MDE 方法的核心关注点)